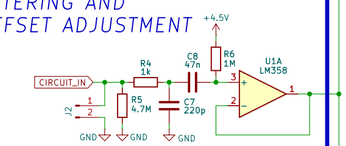
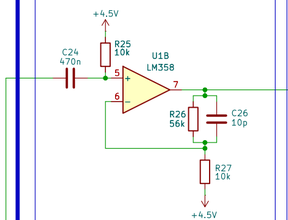
INPUT STAGE: PREFILTERING AND DC-OFFSET ADJUSTMENT



R5: weak pulldown. Pulls the input to ground when no external signal is present.
R4–C7: Passive, Low-Pass filter. Removes unwanted high-frequency noise.
C8: Removes any unwanted DC offset from the input signal.
R6: weak pullup. Injects a stable 4.5V DC offset to the signal
U1B: Voltage buffer. Acts as a low-impedance voltage source for the next stages.

NON-INVERTING PRE-AMPLIFIER WITH LOW-PASS FILTER

U4A acts as a non-inverting amplifier when the input is well below the cutoff frequency of the feedback network. When the input is above the cutoff frequency, it acts as a voltage buffer.



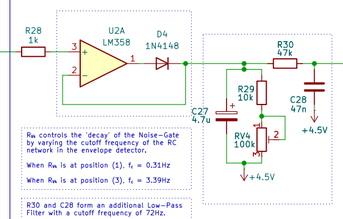
$$H(s) = 1 + \frac{R_{26}}{R_{27}} \cdot \frac{1}{1 + s \cdot R_{26} C_{26}}$$

$$f_c = \frac{1}{2\pi \cdot R_{26} C_{26}}$$

The DC gain of the pre-amplifier is 16.4dB
The cutoff frequency is 284kHz

ENVELOPE DETECTOR

U4B and D4 work together as an ideal diode (ie near-zero threshold voltage)



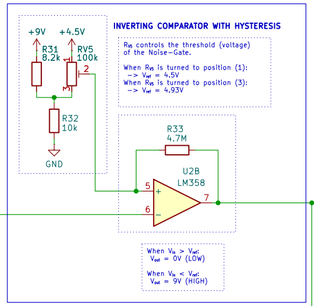
R_{29} controls the 'decay' of the Noise-Gate by varying the cutoff frequency of the RC network in the envelope detector.

When R_{29} is at position (1), f_c = 0.31Hz

When R_{29} is at position (3), f_c = 3.39Hz

R30 and C28 form an additional Low-Pass Filter with a cutoff frequency of 72Hz.

INVERTING COMPARATOR WITH HYSTERESIS



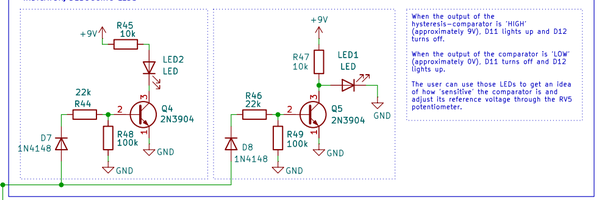
R_{29} controls the threshold (voltage) of the Noise-Gate.

When R_{29} is turned to position (1):
-> V_{ref} = 4.5V
When R_{29} is turned to position (3):
-> V_{ref} = 4.93V

When $V_{in} > V_{ref}$:
V_{out} = 0V (LOW)

When $V_{in} < V_{ref}$:
V_{out} = 9V (HIGH)

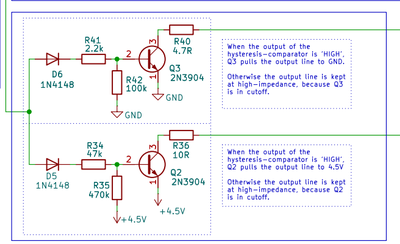
INDICATOR/DEBUGGING LEDS



When the output of the hysteresis-comparator is 'HIGH' (approximately 9V), D11 lights up and D12 turns off.

When the output of the comparator is 'LOW' (approximately 0V), D11 turns off and D12 lights up.

The user can use those LEDs to get an idea of how 'sensitive' the comparator is and adjust its reference voltage through the RV5 potentiometer.



When the output of the hysteresis-comparator is 'HIGH', Q3 pulls the output line to GND.

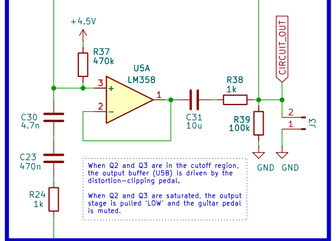
Otherwise the output line is kept at high-impedance, because Q3 is in cutoff.

When the output of the hysteresis-comparator is 'HIGH', Q2 pulls the output line to 4.5V

Otherwise the output line is kept at high-impedance, because Q2 is in cutoff.

NOISE GATE

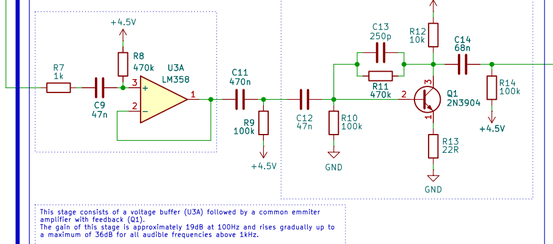
OUTPUT STAGE



When Q2 and Q3 are in the cutoff region, the output buffer (U5B) is driven by the distortion-clipping pedal.

When Q2 and Q3 are saturated, the output stage is pulled 'LOW' and the guitar pedal is muted.

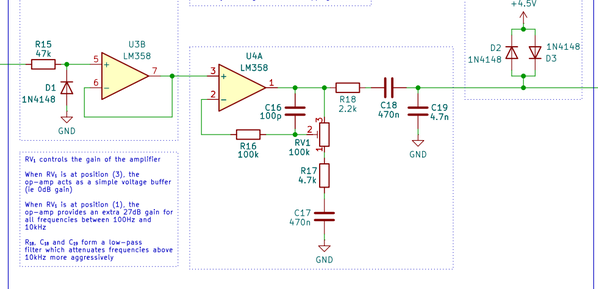
FIRST STAGE AMPLIFIER



This stage consists of a voltage buffer (U3A) followed by a common emmiter amplifier with feedback (Q1).
The gain of this stage is approximately 19dB at 100Hz and rises gradually up to a maximum of 36dB for all audible frequencies above 1kHz.

SECOND STAGE AMPLIFIER AND CLIPPING

This stage consists of a band-pass amplifier with adjustable gain and two clipping diodes.



RV_1 controls the gain of the amplifier

When RV_1 is at position (3), the op-amp acts as a simple voltage buffer (ie 0dB gain)

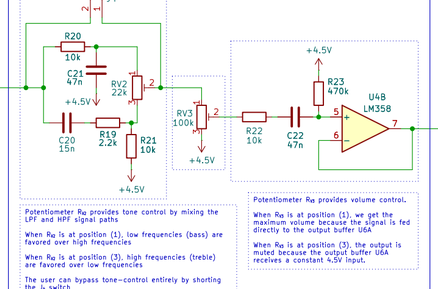
When RV_1 is at position (1), the op-amp provides an extra 27dB gain for all frequencies between 100Hz and 10kHz

R_{18}, C_{18} and C_{19} form a low-pass filter which attenuates frequencies above 10kHz more aggressively

R_{20} and C_{21} form a Low-Pass-Filter (LPF)

C_{20}, R_{19} and R_{21} form a High-Pass-Filter (HPF)

TONE AND VOLUME CONTROL



Potentiometer R_{20} provides tone control by mixing the LPF and HPF signal paths

When R_{20} is at position (1), low frequencies (bass) are favored over high frequencies

When R_{20} is at position (3), high frequencies (treble) are favored over low frequencies

The user can bypass tone-control entirely by shorting the J_4 switch

Potentiometer R_{24} provides volume control.

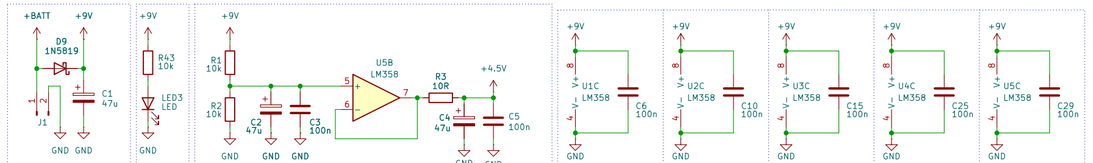
When R_{24} is at position (1), we get the maximum volume because the signal is fed directly to the output buffer U6A

When R_{24} is at position (3), the output is muted because the output buffer U6A receives a constant 4.5V input.

DISTORTION AND CLIPPING

POWER-SUPPLY: DECOUPLING CAPACITORS AND 4.5V RAIL

D9: Reverse-polarity-protection Schottky diode for 9V battery input.
C1: Decoupling capacitor for 9V rail
D10 – R43: Power indicator LED with current limiting resistor
R1 – R2: Resistor network for generating the reference voltage of the 4.5V rail
U1A: Voltage buffer which acts as a low-impedance 4.5V voltage source.
C4 – C5: Decoupling capacitors for the 4.5V rail.





Sheet: /
File: guitar_pedal.kicad_sch

Title:

Size: User | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 1/1